

5(a)(i)	provides return for the signal	B1
	shields signal from noise	B1
5(a)(ii)	e.g. connection between aerial and TV set	B1
5(b)(i)	gain / dB = $10 \lg (P_1 / P_2)$	C1
	$32 = 10 \lg \{P_{\text{MIN}} / (7.6 \times 10^{-6})\}$	A1
	$P_{\text{MIN}} = 0.012 \text{ W}$	
5(b)(ii)	attenuation per unit length = $(1 / L) \times 10 \lg (P_1 / P_2)$	C1
	$6.3 = (1 / L) \times 10 \lg (2.6 / 0.012)$	
	$L = 3.7 \text{ km}$	A1